

# Notes: Ozbas & Scharfstein (RFS, 2010)

## Evidence on the Dark Side of Internal Capital Markets

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# 1 Big picture

- **Question.** Do diversified conglomerates allocate capital across divisions as efficiently as comparable stand-alone firms, or do internal capital markets distort investment away from high-value opportunities?
- **Core comparison.** The paper compares the investment behavior of stand-alone firms to the investment behavior of *unrelated segments* inside conglomerates.
- **Main outcome.** Investment is measured as segment capital expenditures divided by segment sales.
- **Main opportunity measure.** Investment opportunities are measured by lagged industry Tobin's  $Q$ , calculated as the median bounded  $Q$  of stand-alone firms in the same input-output industry.
- **Main finding.** Stand-alone firms have higher investment sensitivity to industry  $Q$  than unrelated conglomerate segments. In high- $Q$  industries, unrelated segments invest less than comparable stand-alone firms.
- **Why this matters.** The result is evidence of a “dark side” of internal capital markets: headquarters does not simply reallocate capital toward the best opportunities; instead, internal politics or agency problems can lead to distorted allocations.
- **Identification idea.** The paper strengthens the comparison by focusing on unrelated segments, measuring relatedness carefully with BEA input-output data, and matching unrelated segments to stand-alone firms by industry, year, size, age, and profitability.
- **Agency mechanism.** The investment distortion is stronger when top management has low ownership stakes. This suggests that the distorted allocation is related to agency problems at headquarters rather than purely to technology, industry composition, or accounting noise.
- **Economic takeaway.** Internal capital markets can reallocate resources, but they are not automatically efficient. Diversification can weaken investment responsiveness to investment opportunities, especially when managerial incentives are weak.

## 2 Incremental contribution of the paper

### 2.1 Contribution relative to the internal-capital-market literature

The paper contributes to the literature on conglomerates and internal capital markets by making the stand-alone-versus-segment comparison more credible and by linking differences in capital allocation to managerial incentives.

- **Contribution 1: measuring true unrelatedness.** Earlier work often classifies segments as unrelated using two-digit SIC codes. Ozbas and Scharfstein argue that this can misclassify vertically related businesses as unrelated and can misclassify unrelated subindustries as related. They instead use BEA Input-Output Benchmark Surveys to identify within-firm industry relatedness.
- **Contribution 2: focusing on unrelated segments.** Related segments can have transfer pricing, joint production, and co-investment that make segment accounting hard to interpret. The paper deliberately focuses on unrelated segments, where segment-level investment and profitability are less contaminated by internal transactions.
- **Contribution 3: nonparametric matching.** Rather than relying only on linear controls, the paper uses matching estimators to compare unrelated segments to stand-alone firms that are close in industry, year, size, age, and profitability. This avoids relying heavily on extrapolation when covariate distributions differ.
- **Contribution 4: agency-based heterogeneity.** The paper appears to be among the first to show that internal-capital-market inefficiency is more pronounced when top managers have low ownership stakes. This directly connects the empirical pattern to agency theories of internal capital allocation.

- **Contribution 5: separating underinvestment from stand-alone overinvestment.** Lower  $Q$ -sensitivity of conglomerate segments could mean conglomerate segments underinvest or stand-alone firms overinvest. The managerial-ownership split supports the view that inefficient resource allocation happens inside conglomerates.
- **Contribution 6: bridging corporate finance and organizational economics.** The paper treats organizational form as a determinant of real investment decisions, connecting corporate diversification, managerial incentives, internal politics, and capital budgeting.

**Incremental contribution in one sentence.** The paper improves measurement, matching, and mechanism testing in the internal-capital-market literature and shows that inefficient capital allocation is especially visible in unrelated conglomerate segments when top-management incentives are weak.

## 3 Data and measurement

### 3.1 Compustat segment data

- The data come from Compustat segment files covering 1979–2006.
- Segment-level accounting variables include sales, assets, capital expenditures, operating profits, depreciation, and industry classification codes.
- The effective regression sample starts in 1980 because lagged industry  $Q$  is needed.
- Segment records are cross-validated against annual firm-level files; observations are dropped when total segment sales do not fall within 25 percent of total firm sales.
- Segments are dropped if they are named “other,” have primary SIC code zero, have incomplete data, have anomalous accounting values, or have real sales below 20 million in 1982 dollars.
- Regulated industries are excluded: transportation, telecommunications, utilities, banking, and insurance.

**Interpretation:** These filters are designed to make the segment-level accounting data usable for comparing real investment across organizational forms. The central concern is that segment data can be noisy, especially when divisions are related through internal transactions.

### 3.2 Stand-alone firms versus unrelated conglomerate segments

- **Stand-alone firms** are single-segment firms.
- **Unrelated segments** are segments of diversified conglomerates that are not related to any other segment within the same firm.
- The paper focuses on unrelated segments because the theories of inefficient internal capital markets are sharpest for diversified firms, and because unrelated divisions are less likely to be affected by transfer pricing and co-investment.

**Interpretation:** The cleanest test of internal-capital-market distortion asks whether a division that looks like an independent business invests differently simply because it is housed inside a diversified firm.

### 3.3 Input-output relatedness measure

- The paper uses six BEA Input-Output Benchmark Surveys: 1977, 1982, 1987, 1992, 1997, and 2002.
- A pair of industries is vertically related if one industry buys more than 10 percent of its inputs from the other or sells more than 10 percent of its outputs to the other.

- Two segments are also considered related if they operate in the same input-output industry.
- A segment is classified as unrelated only if it is not related to any other segment within the same firm.

**Interpretation:** This improves on two-digit SIC relatedness because SIC codes are organized horizontally and can miss vertical relationships. For example, oil extraction and refining can be economically related even if they have different two-digit SIC codes.

### 3.4 Main variables

- Investment outcome:

$$cx_{i,j,t} = \frac{\text{capital expenditures}_{i,j,t}}{\text{sales}_{i,j,t}}.$$

- Cash flow control:

$$cfs_{i,j,t} = \frac{\text{operating profits}_{i,j,t} + \text{depreciation}_{i,j,t}}{\text{sales}_{i,j,t}}.$$

- Industry investment opportunity:

$$Q_{j,t-1} = \text{median bounded Tobin's } Q \text{ of stand-alone firms in industry } j.$$

- Managerial ownership:

$$MO_{i,t} = \frac{\text{stocks and options held by top officers}}{\text{shares outstanding}}.$$

**Interpretation:** Capital spending over sales is a scale-free segment investment rate. Industry  $Q$  is a proxy for investment opportunities that should increase investment if capital is allocated efficiently. Managerial ownership is a proxy for top management incentives.

## 4 Models and empirical specifications

### 4.1 Baseline investment sensitivity regression

The baseline panel analysis estimates variants of:

$$cx_{i,j,t} = \alpha + \beta Q_{j,t-1} + \gamma(Q_{j,t-1} \times Unrelated_i) + \delta cfs_{i,j,t} + \eta(cfs_{i,j,t} \times Unrelated_i) + \kappa Unrelated_i + FE + \varepsilon_{i,j,t}. \quad (1)$$

- $\beta$  measures the  $Q$ -sensitivity of investment for stand-alone firms.
- $\gamma$  measures the difference in  $Q$ -sensitivity between unrelated segments and stand-alone firms.
- A negative  $\gamma$  means unrelated segments respond less to investment opportunities.
- The paper uses industry fixed effects, year fixed effects, and in some specifications segment fixed effects.

**Interpretation:** The central empirical test is whether conglomerate segments invest less when their industry opportunities improve. If internal capital markets work well, unrelated segments should be at least as responsive to  $Q$  as stand-alone firms. The paper finds the opposite.

## 4.2 Matched-pair difference specification

The paper motivates matching with the following general structure:

$$cxs_{j,t}^U = a^U + b_{j,t}(\text{observable}) + [c^U + c_{j,t}(\text{observable})]Q_{j,t-1} + dcfs_t^U, \quad (2)$$

$$cxs_{j,t}^S = a^S + b_{j,t}(\text{observable}) + [c^S + c_{j,t}(\text{observable})]Q_{j,t-1} + dcfs_t^S. \quad (3)$$

For matched pairs, the difference removes the common observable industry-year effects:

$$\Delta cxs_t = (a^U - a^S) + (c^U - c^S)Q_{j,t-1} + d(cfs_t^U - cfs_t^S) + \Delta \varepsilon_t. \quad (4)$$

**Interpretation:** Matching is intended to remove unknown nonlinear industry-year, size, age, and profitability effects. The coefficient on  $Q$  in the difference regression measures whether unrelated segments have lower  $Q$ -sensitivity than matched stand-alone firms.

## 4.3 Bias-corrected matching estimator

The paper also uses the Abadie-Imbens bias-corrected matching estimator. Conceptually, the estimand is:

$$ATT = E[Y^U - Y^S(0) \mid U = 1], \quad (5)$$

where  $Y^U$  is the investment outcome of an unrelated segment and  $Y^S(0)$  is the counterfactual investment outcome if that same business were instead operated as a matched stand-alone firm.

- Matching is exact on industry and year.
- Matching is continuous on sales, age, profitability, and in the agency section managerial ownership.
- The number of matches is four.
- Treatment effects are estimated separately for low- $Q$  and high- $Q$  bins.

**Interpretation:** Bias-corrected matching addresses the concern that linear regressions compare segments and stand-alone firms that are not really comparable. The key question is whether high- $Q$  unrelated segments still invest less after this comparison is made local and nonparametric.

## 4.4 Agency heterogeneity specification

The agency test estimates differences between matched pairs as a function of managerial ownership:

$$\Delta cxs_t = \alpha + \beta Q_{j,t-1} + \rho MO_{i,t} + \theta(Q_{j,t-1} \times MO_{i,t}) + \delta \Delta cfs_t + \varepsilon_t. \quad (6)$$

- $\beta < 0$  means that, when managerial ownership is low, unrelated segments are less  $Q$ -sensitive than stand-alone firms.
- $\theta > 0$  means higher managerial ownership mitigates the lower  $Q$ -sensitivity of unrelated segments.
- The theory-driven prediction is that agency problems should be more severe when managerial ownership is low.

**Interpretation:** The managerial-ownership test is the paper's key mechanism test. If the investment distortion were due only to technology or measurement, it should not naturally vary with top-management ownership.

## 5 Identification and empirical design

### 5.1 Core identification problem

The empirical challenge is that unrelated conglomerate segments are not randomly assigned to be inside conglomerates. They may differ from stand-alone firms in size, age, profitability, industry composition, and accounting characteristics.

- Conglomerate segments may be larger than stand-alone firms.
- Conglomerate segments may operate in industries with different natural investment sensitivities.
- Older firms may have different growth opportunities from younger firms.
- Related segments may have transfer pricing and co-investment that contaminate segment accounting.

**Interpretation:** The paper’s empirical design is not a randomized experiment. It instead builds a sequence of comparisons that progressively removes plausible confounds.

### 5.2 Why the unrelated-segment sample is useful

Focusing on unrelated segments serves two purposes:

- It maps closely to theories of diversified internal capital markets, where headquarters allocates resources across unrelated businesses.
- It reduces measurement error from within-firm vertical transactions and co-specialized investment.

**Interpretation:** If the paper compared all segments to stand-alone firms, differences could be driven by transfer pricing or joint investment in vertically related divisions. The unrelated filter makes the investment comparison cleaner.

### 5.3 Why matching matters

The matching design compares each unrelated segment to stand-alone firms that are in the same industry and year and similar in size, age, and profitability.

- Exact industry-year matching controls for time-varying industry investment opportunities and shocks.
- Size matching addresses adjustment-cost and scale differences.
- Age matching addresses life-cycle differences.
- Profitability matching addresses internal cash-flow differences.

**Interpretation:** Matching lets the paper compare a conglomerate segment to a stand-alone firm facing similar observable circumstances. The result that high- $Q$  unrelated segments still invest less is therefore harder to explain by observable composition.

### 5.4 Why managerial ownership is informative

Managerial ownership is not randomly assigned, but it is informative as a mechanism proxy.

- If low  $Q$ -sensitivity were purely due to technology, matching, or segment accounting, it should not disappear in high-managerial-ownership firms.

- If weak headquarters incentives drive inefficient resource allocation, the pattern should be stronger when managerial ownership is low.
- The paper finds exactly this: the low- $Q$ -sensitivity/high- $Q$  underinvestment pattern is strongest in low managerial ownership conglomerates.

**Interpretation:** The agency evidence is a mechanism test rather than causal identification of managerial ownership. It strengthens the interpretation that internal capital market distortions are tied to agency problems.

## 5.5 Main limitations

- The paper does not have exogenous variation in diversification or managerial ownership.
- Segment-level accounting data may still contain measurement error.
- Industry  $Q$  is an imperfect proxy for segment-specific investment opportunities.
- Managerial ownership could proxy for other unobserved characteristics.
- The study focuses on capital expenditures, not R&D, marketing, pricing, or intangible investments.

**Interpretation:** The paper's strongest claim is not that it has a natural experiment. Its strength is that several increasingly refined comparisons point to the same pattern, and the pattern lines up with agency theories of internal capital markets.

## 6 Section-by-section study guide

- **Introduction.** Frames the internal capital market debate: the bright side is better information and capital redeployment; the dark side is divisional rent-seeking and inefficient cross-subsidization.
- **Data section.** Explains how the paper constructs the sample of stand-alone firms and unrelated conglomerate segments and why relatedness measurement is central.
- **Panel analysis.** Shows the baseline fact: unrelated segments have lower  $Q$ -sensitivity than stand-alone firms.
- **Matching analysis.** Shows that the baseline fact survives matching on industry, year, size, age, and profitability.
- **Agency evidence.** Shows that the distortion is more pronounced when top management has lower ownership, supporting agency-based theories.
- **Conclusion.** Interprets the evidence as suggestive of inefficient internal capital markets while emphasizing the need to understand lower-level agency problems and organizational processes.

## 7 Tables and figures

### 7.1 Table 1: Descriptive statistics

**Read:**

- Table 1 compares stand-alone firms and unrelated conglomerate segments in the full sample and in manufacturing industries.



- In the full sample, unrelated segments are larger: mean segment sales are 1,150 versus 779 for stand-alone firms, and mean segment assets are 932 versus 773.
- Unrelated segments have more capital expenditures in dollars, but a lower investment rate: capital expenditures/sales are 0.061 versus 0.072.
- Unrelated segments have lower industry investment opportunities: lagged industry  $Q$  is 1.31 versus 1.42 for stand-alone firms.
- The manufacturing sample displays the same broad pattern: unrelated segments are larger but have lower capex/sales and lower lagged industry  $Q$ .

**Detailed interpretation:** Table 1 establishes the central selection issue. Unrelated segments are not simply smaller versions of stand-alone firms; they are larger and operate in somewhat lower- $Q$  industries. This is why simple comparisons are not enough. The table motivates the later use of controls and matching. It also foreshadows the paper’s main result: despite higher absolute capital expenditures, unrelated segments invest less as a percentage of sales and appear less exposed to strong investment opportunities.

**Economic message:** The descriptive evidence does not yet prove inefficient capital allocation, but it shows why a careful empirical strategy is necessary. Organizational form is correlated with scale and industry opportunity, so the paper must compare like with like.

**Table 1**  
Descriptive statistics

| Sample: All industries            | Stand-alone |       | Unrelated |       |
|-----------------------------------|-------------|-------|-----------|-------|
| Segment level                     | Mean        | SD    | Mean      | SD    |
| Segment sales                     | 779         | 3,778 | 1,150***  | 6,117 |
| Segment assets                    | 773         | 4,530 | 932***    | 4,483 |
| Segment capital expenditure       | 44          | 223   | 68***     | 413   |
| Segment cash flow                 | 99          | 486   | 154***    | 740   |
| Segment capital expenditure/sales | 0.072       | 0.116 | 0.061***  | 0.098 |
| Segment cash flow/sales           | 0.121       | 0.158 | 0.139***  | 0.135 |
| Lagged industry $Q$               | 1.42        | 0.40  | 1.31***   | 0.37  |
| Obs                               | 61,081      |       | 13,186    |       |

  

| Sample: Manufacturing industries  | Stand-alone |       | Unrelated |       |
|-----------------------------------|-------------|-------|-----------|-------|
| Segment level                     | Mean        | SD    | Mean      | SD    |
| Segment sales                     | 720         | 3,299 | 1,271***  | 6,909 |
| Segment assets                    | 700         | 3,301 | 990***    | 4,989 |
| Segment capital expenditure       | 46          | 263   | 72***     | 461   |
| Segment cash flow                 | 103         | 515   | 166***    | 815   |
| Segment capital expenditure/sales | 0.059       | 0.072 | 0.046***  | 0.048 |
| Segment cash flow/sales           | 0.108       | 0.124 | 0.126***  | 0.097 |
| Lagged industry $Q$               | 1.43        | 0.41  | 1.29***   | 0.35  |
| Obs                               | 30,645      |       | 9,978     |       |

Descriptive statistics

**Table 2**  
 $Q$ -sensitivity of investment: Unrelated segments and stand-alone firms

| Sample                               | All (1)              | Manufacturing (2)    | All (3)              | Manufacturing (4)    |
|--------------------------------------|----------------------|----------------------|----------------------|----------------------|
| Lagged industry $Q$                  | 0.025***<br>[0.003]  | 0.025***<br>[0.002]  | 0.026***<br>[0.003]  | 0.023***<br>[0.003]  |
| Lagged industry $Q \times Unrelated$ | -0.017***<br>[0.003] | -0.016***<br>[0.003] | -0.013***<br>[0.003] | -0.010***<br>[0.003] |
| Cash flow/sales                      | 0.115***<br>[0.014]  | 0.066***<br>[0.013]  | 0.007<br>[0.010]     | -0.009<br>[0.009]    |
| Cash flow/sales $\times Unrelated$   | -0.028*<br>[0.016]   | 0.017<br>[0.014]     | -0.037<br>[0.024]    | -0.004<br>[0.016]    |
| Unrelated                            | 0.017***<br>[0.005]  | 0.010***<br>[0.003]  | 0.017***<br>[0.005]  | 0.012**<br>[0.005]   |
| Industry F.E.                        | Yes                  | Yes                  | No                   | No                   |
| Segment F.E.                         | No                   | No                   | Yes                  | Yes                  |
| Year F.E.                            | Yes                  | Yes                  | Yes                  | Yes                  |
| $R^2$                                | 0.385                | 0.101                | 0.749                | 0.611                |
| Obs                                  | 74,267               | 40,623               | 74,267               | 40,623               |

Unrelated segments and stand-alone firms are compared (Compustat segment files, 1980–2006). Dependent variable is capital spending over sales. Industry definitions follow the Input-Output Benchmark Surveys of the Bureau of Economic Analysis. Industry  $Q$  in a given year is the median bounded  $Q$  of stand-alone firms in the industry. Columns 2 and 4 restrict the sample to manufacturing industries. Heteroskedasticity-robust standard errors are in brackets. Standard errors are reported for clusters at the industry-year level. \* statistic is linear.

$Q$ -sensitivity of investment: unrelated segments and stand-alone firms

## 7.2 Table 2: $Q$ -sensitivity of investment: unrelated segments and stand-alone firms

Read:

- The coefficient on lagged industry  $Q$  is positive and significant, meaning stand-alone firms invest more when industry investment opportunities improve.
- The interaction  $Q \times Unrelated$  is negative and significant in all specifications.
- In the full sample, the interaction is about  $-0.017$  or  $-0.013$  depending on fixed effects.
- In manufacturing industries, the interaction is about  $-0.016$  or  $-0.010$ .
- The economic interpretation in the text is that when  $Q$  rises from 1.40 to 1.80, stand-alone investment rises much more than unrelated-segment investment.

**Detailed interpretation:** Table 2 is the baseline “dark side” result. Conglomerate segments do invest in response to  $Q$ , but less than stand-alone firms. The negative interaction captures a slope difference, not just a level difference. This matters because efficient internal capital markets should direct capital toward businesses with better opportunities. The table also includes cash-flow controls and alternative fixed effects, showing that the lower  $Q$ -sensitivity is not only a firm-level or industry-level difference.

**Economic message:** If  $Q$  proxies for growth opportunities, then lower  $Q$ -sensitivity means conglomerate headquarters does not push capital toward high-opportunity unrelated divisions as aggressively as external markets push stand-alone firms.

### 7.3 Table 3: Difference between pairs of unrelated segments and stand-alone firms: industry, size, and age matched

**Read:**

- The dependent variable is the investment difference within a matched pair: unrelated segment minus stand-alone firm.
- Matching is exact on industry and year.
- Columns add matching on sales, sales plus age, and sales plus broader age categories.
- The lagged industry  $Q$  coefficient is negative and significant in all panels.
- The pattern holds in all industries, manufacturing industries, and alternative relatedness-threshold samples.

**Detailed interpretation:** Table 3 answers the concern that Table 2 is driven by differences in observable characteristics. Matched pairs are designed to face the same industry-year investment environment, so the negative  $Q$  coefficient in the difference regression says that even within comparable pairs, the conglomerate segment invests less than the stand-alone firm when industry  $Q$  is higher. The alternative relatedness threshold checks that the result is not an artifact of the 10 percent input-output cutoff.

**Economic message:** The matching evidence makes the result more credible: the low investment responsiveness of unrelated segments is not simply because they are in different industries, have different sizes, or differ in age.

### 7.4 Table 4: Bias-corrected matching estimates

**Read:**

- Table 4 estimates average treatment effects for unrelated segments relative to matched stand-alone firms.
- In high- $Q$  industries, unrelated segments invest less than matched stand-alone firms in every specification.
- The high- $Q$  effects range from about  $-0.0031$  to  $-0.0097$ .
- In low- $Q$  industries, estimates are mixed and usually insignificant.
- The high-minus-low difference is always negative and significant.

**Detailed interpretation:** Table 4 uses a nonparametric bias-corrected matching estimator rather than a regression slope. It asks: for a treated unrelated segment, what is the average investment difference relative to similar stand-alone control firms? The answer is clearest in high- $Q$  industries: unrelated segments underinvest relative to comparable stand-alone firms. The H-L difference summarizes the allocative distortion: the investment gap is more negative when investment opportunities are strong.

**Table 3**  
Difference between pairs of unrelated segments and stand-alone firms: Industry, size, and age matched

|                                                                           | (1)                  | (2)                  | (3)                  |
|---------------------------------------------------------------------------|----------------------|----------------------|----------------------|
| Panel A: All industries                                                   |                      |                      |                      |
| Constant                                                                  | 0.021***<br>[0.005]  | 0.021***<br>[0.006]  | 0.014**<br>[0.006]   |
| Lagged industry $Q$                                                       | -0.022***<br>[0.004] | -0.019***<br>[0.004] | -0.012***<br>[0.004] |
| Difference in cash flow/sales                                             | 0.117***<br>[0.017]  | 0.109***<br>[0.019]  | 0.138***<br>[0.021]  |
| $R^2$                                                                     | 0.029                | 0.024                | 0.036                |
| Obs                                                                       | 9,176                | 6,001                | 4,282                |
| Panel B: Manufacturing industries                                         |                      |                      |                      |
| Constant                                                                  | 0.013**<br>[0.005]   | 0.012***<br>[0.004]  | 0.017***<br>[0.006]  |
| Lagged industry $Q$                                                       | -0.017***<br>[0.004] | -0.015***<br>[0.003] | -0.018***<br>[0.004] |
| Difference in cash flow/sales                                             | 0.091***<br>[0.014]  | 0.089***<br>[0.014]  | 0.075***<br>[0.016]  |
| $R^2$                                                                     | 0.029                | 0.031                | 0.030                |
| Obs                                                                       | 6,904                | 4,130                | 2,789                |
| Panel C: All industries, alternative relatedness threshold (5%)           |                      |                      |                      |
| Constant                                                                  | 0.035***<br>[0.006]  | 0.033***<br>[0.008]  | 0.031***<br>[0.009]  |
| Lagged industry $Q$                                                       | -0.030***<br>[0.004] | -0.027***<br>[0.005] | -0.026***<br>[0.006] |
| Difference in cash flow/sales                                             | 0.060***<br>[0.019]  | 0.056**<br>[0.028]   | 0.087***<br>[0.029]  |
| $R^2$                                                                     | 0.016                | 0.013                | 0.023                |
| Obs                                                                       | 4,995                | 3,046                | 2,098                |
| Panel D: Manufacturing industries, alternative relatedness threshold (5%) |                      |                      |                      |
| Constant                                                                  | 0.022***<br>[0.007]  | 0.021***<br>[0.005]  | 0.027***<br>[0.007]  |
| Lagged industry $Q$                                                       | -0.023***<br>[0.005] | -0.021***<br>[0.004] | -0.024***<br>[0.006] |
| Difference in cash flow/sales                                             | 0.053***<br>[0.014]  | 0.071***<br>[0.016]  | 0.065***<br>[0.021]  |
| $R^2$                                                                     | 0.020                | 0.033                | 0.036                |
| Obs                                                                       | 3,972                | 2,256                | 1,499                |

Unrelated segments are matched with stand-alone firms (Compustat segment files, 1980–2006). In column 1, unrelated segments are matched with stand-alone firms based on industry and year. In column 2, unrelated segments are matched with stand-alone firms based on industry, year, and sales. In column 3, unrelated segments

table3\_matched\_difference.png

Difference between pairs of unrelated segments and stand-alone firms: industry, size, and age matched

**Economic message:** This table is the paper’s strongest robustness check on the main result. It shows that the dark-side pattern survives a local comparison and does not rely on global regression functional form.

## 7.5 Table 5: Evidence on agency: Difference between matched pairs of unrelated segments and stand-alone firms

**Read:**

- Table 5 regresses the matched-pair investment difference on industry  $Q$ , managerial ownership, and the interaction between the two.
- The lagged industry  $Q$  coefficient is negative and significant.
- The interaction  $Q \times$  managerial ownership is positive and significant in most specifications.
- The level effect of managerial ownership is negative, while the interaction offsets the negative  $Q$  sensitivity at higher ownership levels.
- The pattern is present in both the full sample and the manufacturing sample.

**Detailed interpretation:** Table 5 links inefficient allocation to agency. If headquarters incentives are weak, unrelated segments exhibit a larger low- $Q$ -sensitivity pattern. Higher managerial ownership makes conglomerate investment differences look more like stand-alone investment. This supports the Scharfstein-Stein agency view in which both top-level and divisional incentives matter for inefficient internal capital allocation.

**Economic message:** Managerial ownership does not merely affect firm value; it affects real capital budgeting. Stronger top-management incentives appear to mitigate the dark side of internal capital markets.

**Table 4**  
Bias-corrected matching estimates

| Match variables                   | Lagged industry $Q$    |                        | Difference H-L         |
|-----------------------------------|------------------------|------------------------|------------------------|
|                                   | Low                    | High                   |                        |
| Panel A: All industries           |                        |                        |                        |
| <i>Sales, profitability</i>       | -0.0025*<br>[0.0015]   | -0.0086***<br>[0.0016] | -0.0060***<br>[0.0022] |
| <i>Age, profitability</i>         | 0.0024<br>[0.0016]     | -0.0034**<br>[0.0016]  | -0.0058***<br>[0.0022] |
| <i>Sales, age, profitability</i>  | 0.0030*<br>[0.0016]    | -0.0031**<br>[0.0016]  | -0.0061***<br>[0.0022] |
| Panel B: Manufacturing industries |                        |                        |                        |
| <i>Sales, profitability</i>       | -0.0034***<br>[0.0012] | -0.0097***<br>[0.0012] | -0.0063***<br>[0.0017] |
| <i>Age, profitability</i>         | -0.0007<br>[0.0011]    | -0.0058***<br>[0.0012] | -0.0051***<br>[0.0016] |
| <i>Sales, age, profitability</i>  | -0.0004<br>[0.0011]    | -0.0054***<br>[0.0012] | -0.0050***<br>[0.0016] |

Abadie and Imbens (2007) bias-corrected estimates for the average treatment effect for treated unrelated segments relative to control stand-alone firms (Compustat segment files, 1980–2006). Treatment outcome is capital spending over sales ratio. Matching is continuous with respect to sales, age, and profitability (cash flow over sales ratio) and exact with respect to industry and year. Number of matches is four. Low- and high- $Q$  bins are based on the annual sample median of lagged industry  $Q$ . Panel B restricts the sample to manufacturing industries. Standard errors are in brackets. Comparisons between low- and high- $Q$  bins assume independence of estimated average treatment effects. Asterisks indicate significance at the 10% (\*), 5% (\*\*), and 1% (\*\*\*) levels.

Bias-corrected matching estimates

**Table 5**  
Evidence on agency: Difference between matched pairs of unrelated segments and stand-alone firms

|                                                 | (1)                  | (2)                  | (3)                  |
|-------------------------------------------------|----------------------|----------------------|----------------------|
| Panel A: All industries                         |                      |                      |                      |
| Constant                                        | 0.045***<br>[0.010]  | 0.070***<br>[0.018]  | 0.059***<br>[0.020]  |
| Lagged industry $Q$                             | -0.035***<br>[0.006] | -0.050***<br>[0.011] | -0.040***<br>[0.011] |
| Management ownership                            | -0.220**<br>[0.096]  | -0.296**<br>[0.141]  | -0.349**<br>[0.170]  |
| Lagged industry $Q \times$ Management ownership | 0.142**<br>[0.059]   | 0.197**<br>[0.091]   | 0.243**<br>[0.108]   |
| Difference in cash flow/sales                   | 0.094***<br>[0.031]  | 0.128***<br>[0.044]  | 0.124**<br>[0.054]   |
| $R^2$                                           | 0.031                | 0.049                | 0.040                |
| Obs                                             | 2,349                | 1,495                | 1,211                |
| Panel B: Manufacturing industries               |                      |                      |                      |
| Constant                                        | 0.020**<br>[0.008]   | 0.015<br>[0.010]     | 0.025**<br>[0.012]   |
| Lagged industry $Q$                             | -0.022***<br>[0.005] | -0.017**<br>[0.007]  | -0.022***<br>[0.008] |
| Management ownership                            | -0.163**<br>[0.079]  | -0.163<br>[0.165]    | -0.429*<br>[0.237]   |
| Lagged industry $Q \times$ Management ownership | 0.109**<br>[0.051]   | 0.110<br>[0.109]     | 0.278*<br>[0.160]    |
| Difference in cash flow/sales                   | 0.070***<br>[0.027]  | 0.149***<br>[0.027]  | 0.112***<br>[0.033]  |
| $R^2$                                           | 0.023                | 0.092                | 0.076                |
| Obs                                             | 1,681                | 1,012                | 764                  |

Unrelated segments are matched with stand-alone firms (Compustat segment files, 1980–2006). The matching procedure is described in Table 3. Dependent variable is the difference in the capital spending over sales ratio of the matched pair, unrelated segment minus stand-alone firm. (Using Standard & Poor’s ExecuComp database) Management Ownership is defined as the sum of stocks and options held by top officers as a fraction of outstanding shares. Panel B restricts the sample to manufacturing industries. Heteroskedasticity robust standard

**Table 6**  
Evidence on agency: Bias-corrected matching estimates

| Management ownership              | Lagged industry $Q$   |                        | Difference H-L         |
|-----------------------------------|-----------------------|------------------------|------------------------|
|                                   | Low                   | High                   |                        |
| Panel A: All industries           |                       |                        |                        |
| <i>All</i>                        | 0.0146**<br>[0.0061]  | -0.0085**<br>[0.0035]  | -0.0231***<br>[0.0070] |
| <i>Low</i>                        | 0.0222***<br>[0.0084] | -0.0130***<br>[0.0038] | -0.0352***<br>[0.0092] |
| <i>High</i>                       | 0.0021<br>[0.0100]    | -0.0050<br>[0.0076]    | -0.0071<br>[0.0126]    |
| Panel B: Manufacturing industries |                       |                        |                        |
| <i>All</i>                        | 0.0050<br>[0.0046]    | -0.0159***<br>[0.0057] | -0.0209***<br>[0.0073] |
| <i>Low</i>                        | 0.0090*<br>[0.0049]   | -0.0221***<br>[0.0063] | -0.0311***<br>[0.0079] |
| <i>High</i>                       | -0.0046<br>[0.0090]   | -0.0035<br>[0.0115]    | 0.0011<br>[0.0146]     |

Abadie and Imbens (2007) bias-corrected estimates for the average treatment effect for treated unrelated segments relative to control stand-alone firms (Compustat segment files, 1980–2006). Treatment outcome is capital spending over sales ratio. Matching is continuous with respect to sales, age, management ownership, and profitability (cash flow over sales ratio) and exact with respect to industry and year. Number of matches is four. Low- and high- $Q$  bins are based on the annual sample median of lagged industry  $Q$ . Low- and high-management ownership bins are based on the annual sample median of management ownership. Panel B restricts the sample to manufacturing industries. Standard errors are in brackets. Comparisons between different bins assume independence of estimated average treatment effects. Asterisks indicate significance at the 10% (\*), 5% (\*\*), and 1% (\*\*\*) levels.

Evidence on agency: difference between matched pairs of unrelated segments and stand-alone firms

Evidence on agency: bias-corrected matching estimates

## 7.6 Table 6: Evidence on agency: Bias-corrected matching estimates

Read:

- Table 6 combines the bias-corrected matching design with the managerial-ownership split.
- In all industries, high- $Q$  unrelated segments invest  $-0.0085$  less than matched stand-alone firms; in low managerial ownership firms the effect is  $-0.0130$ .
- In manufacturing industries, the high- $Q$  effect is  $-0.0159$  overall and  $-0.0221$  in low managerial ownership firms.
- In high managerial ownership firms, the effect largely disappears.
- The high-minus-low difference is strongly negative for low managerial ownership but not for high managerial ownership.

**Detailed interpretation:** Table 6 is the cleanest mechanism table because it combines matching and managerial incentive heterogeneity. The underinvestment of unrelated segments in high- $Q$  industries is not just a regression artifact; it survives matching and is concentrated exactly where agency theory predicts it should be strongest. The disappearance of the effect in high managerial ownership firms is central to the paper’s interpretation.

**Economic message:** Internal capital markets are not uniformly inefficient. Their performance depends on organizational incentives. Conglomerates with stronger managerial ownership look less distorted, which connects capital allocation to governance.

## 8 Result map

- **Descriptive selection:** unrelated segments are larger, have lower capex/sales, and operate in lower- $Q$  industries than stand-alone firms.
- **Baseline slope result:** the interaction  $Q \times Unrelated$  is negative, so unrelated segments have lower investment- $Q$  sensitivity.

- **Matched-pair result:** within matched pairs, investment differences become more negative as industry  $Q$  rises.
- **Bias-corrected ATT result:** unrelated segments underinvest relative to matched stand-alone firms in high- $Q$  industries.
- **Agency result:** the distortion is stronger when managerial ownership is low and disappears or weakens when managerial ownership is high.
- **Overall interpretation:** the evidence is consistent with internal capital market inefficiency driven at least partly by agency problems.

## 9 Short summary

- The paper asks whether internal capital markets allocate capital efficiently across unrelated divisions of conglomerates.
- It compares unrelated conglomerate segments to stand-alone firms using Compustat segment data from 1979–2006.
- The paper measures relatedness carefully using BEA input-output surveys rather than relying only on SIC codes.
- Investment is measured as capital spending over sales.
- Investment opportunities are measured by lagged industry  $Q$ .
- Stand-alone firms are more responsive to  $Q$  than unrelated conglomerate segments.
- The gap is driven mainly by unrelated segments investing less than stand-alone firms in high- $Q$  industries.
- The finding survives matching on industry, year, sales, age, and profitability.
- Bias-corrected matching estimates confirm high- $Q$  underinvestment by unrelated segments.
- The pattern is stronger in firms where top management has low ownership.
- The pattern weakens or disappears when managerial ownership is high.
- The evidence supports the dark-side view of internal capital markets: agency problems can distort capital allocation.
- The paper does not prove random assignment of organizational form, but it narrows many important alternative explanations.

## 10 Conclusion

### 10.1 Core conclusion

Ozbas and Scharfstein provide evidence that internal capital markets in diversified conglomerates can allocate capital inefficiently. Stand-alone firms respond more strongly to industry investment opportunities than unrelated segments inside conglomerates. The difference is most visible in high- $Q$  industries, where conglomerate segments invest less than comparable stand-alone firms.

### 10.2 Economic intuition

An efficient internal capital market should move resources toward divisions with the best opportunities. In the paper’s setting, this would mean that unrelated segments in high- $Q$  industries should receive capital and invest at least as aggressively as stand-alone firms in the same industries. Instead, the results show muted

investment responsiveness. This suggests that capital allocation inside conglomerates is shaped not only by opportunity but also by organizational frictions, bargaining, internal politics, and agency problems.

### **10.3 Why the incremental contribution matters**

The paper improves the credibility of the internal-capital-market evidence in three ways. First, the input-output relatedness measure reduces measurement error from related divisions. Second, matching avoids comparing very different stand-alone firms and conglomerate segments. Third, managerial ownership heterogeneity links the investment distortion to agency incentives. Together, these contributions move the literature from documenting a diversification discount toward understanding the organizational mechanism behind inefficient investment.

### **10.4 Governance implication**

The managerial ownership evidence suggests that corporate governance can affect real resource allocation. When top managers have stronger ownership incentives, unrelated segments invest more like comparable stand-alone firms. This implies that governance is not just about payout, acquisitions, or valuation; it also affects how firms internally allocate capital across divisions.

### **10.5 Limitations**

The paper cannot fully solve the selection problem because diversification and managerial ownership are endogenous. Industry  $Q$  is an imperfect proxy for division-specific investment opportunities, and segment-level accounting may still be noisy. The paper also focuses on capital expenditures, leaving other forms of investment such as R&D, advertising, pricing policy, and human capital for future work.

### **10.6 Takeaway**

The paper's enduring takeaway is that internal capital markets have both a bright side and a dark side. They may overcome external financing frictions, but they can also misallocate capital when agency problems and divisional rent-seeking distort the budgeting process. The evidence favors the view that the dark side is economically meaningful, especially in unrelated conglomerates with weak managerial incentives.